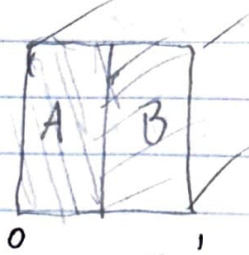


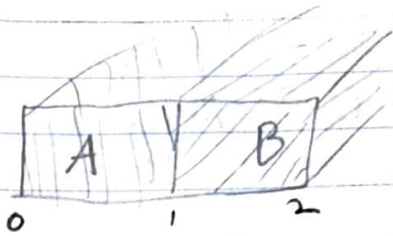
Lecture 11-12

→ Stephen Smale:

12.1: Danish Pastry

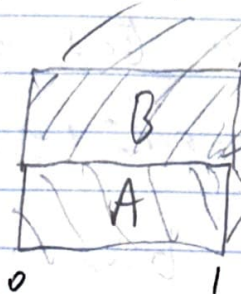


press


$$I_\lambda$$

$$0 \leq \lambda \leq \frac{1}{2}$$

(let $\lambda = \frac{1}{2}$ for now)



cut
place

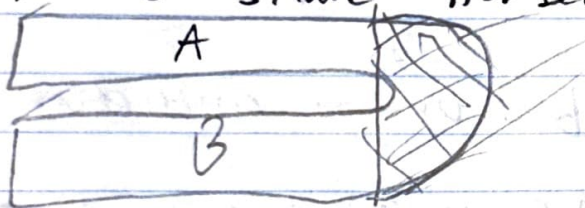
→ repeat...

→ A 3-D object

→ The bakers map

→ generalization / extension of Bernoulli map (1-D chaotic system)

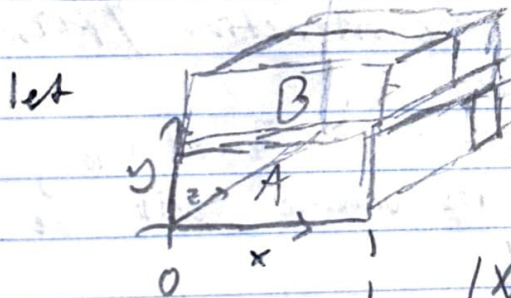
→ Now, consider a 'smale' version of the Baker's map: The Smale Horseshoe (he had the insight)



e had the insight in
Copacabana, now is
the 'grandfather' of
chaotic dynamics in
applied math)

↑ cut off roll-over

→ piecewise-linear dynamics: the system is as nonlinear as possible for what we can do



Now, let us write down the formation for the Baker's map:

Map: if $x_n < 1/2$

$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} 2x_n \\ ay_n \end{pmatrix} \quad \text{if } x_n < 1/2$$

if $x_n > 1/2$

$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} 2x_n - 1 \\ ay_n + 1/2 \end{pmatrix} \quad \text{if } x_n > 1/2$$

← 'B' rests on top of 'A'

shrinks by some factor < 1

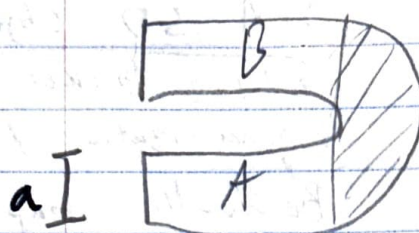
→ Moore, Newton: "if you push something, it pushes back w/ oscillatory motion"

→ We can consider this at various stages of sophistication

→ but what about the other situation, where we cannot treat the response as "negative": you push a cart, it runs away. But it is still limited to the room

→ QFT / other fields have nonlinear coupling taking it a step further

→ Now consider Smale's Map (foldover):



$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = \begin{pmatrix} 1 - ax_n^2 + y_n \\ bx_n \end{pmatrix}$$

$$\Rightarrow J_n = \begin{pmatrix} -2ax_n & 1 \\ b & 0 \end{pmatrix} \quad |b| < 1$$

det $J = -b$: uniform contraction of mapping

Tr $J = -2ax_n$: a large $\rightarrow$ stretching mechanism

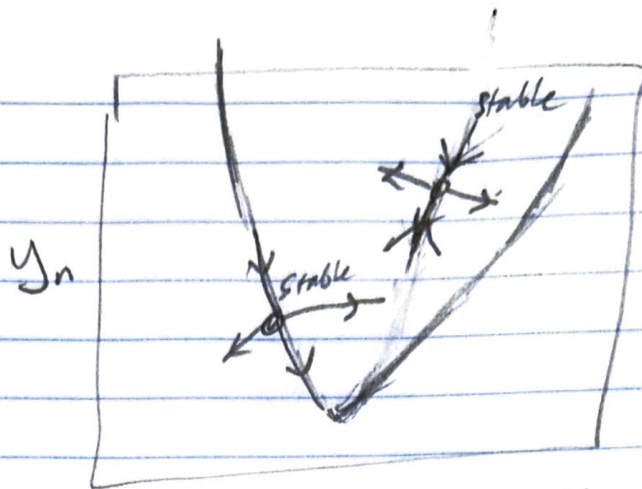
→ By viewing the Jacobian J of the mapping, we can determine the essence of the volumetric transformation, i.e., contraction / mixing

→ Consider a small volume element: by the time the map is applied, its volume will be smaller by a factor of b

Q: how can all volume elements get smaller infinitely if total volume stays constant?

Hénon Map (Pameau)

→ Consider $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y + 1 - ax^2 \end{pmatrix}$, a 2-D mapping representing the folding & Smale's



→ 2 fixed points exist, both are saddles

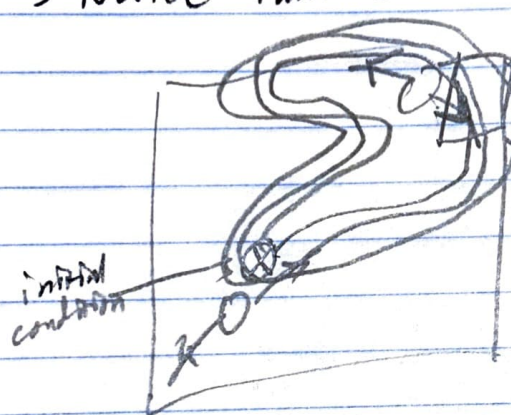
→ $a = 1.4$

$b = 0.3$

x_n → i.e., "not too" stretching

→ Model of a 'typical' chemical/biological nonlinear dynamical system or contracting

→ Notice that orbits behave as fractals:



Fractal!

~~Self~~ - Similar

→ Not a self-similar fractal; is this a strange chaotic system?

→ chaotic systems are nearly impossible for find & prove
↳ a very special case → not a trivial thing